7phish

A Phishing Simulator

Abstract

This research paper introduces 7phish, a simple phishing tool for performing phishing simulations. A simple and effective method to generate various phishing URLs is provided by 7phish. Users may quickly create and set up phishing campaigns using its user-friendly interface to meet their unique needs. Organisations can evaluate their sensitivity to such threats and educate employees on phishing attempts using 7phish. Because of the tool's ease of use and user-friendliness, organisations can proactively increase their cybersecurity defences through focused training and awareness initiatives. 7phish is an excellent tool for improving overall security posture and defending against phishing attacks due to its flexible features and ease of use.

Keywords – *Phishing, Simulator, 7phish*

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Introduction

Phishing Attacks

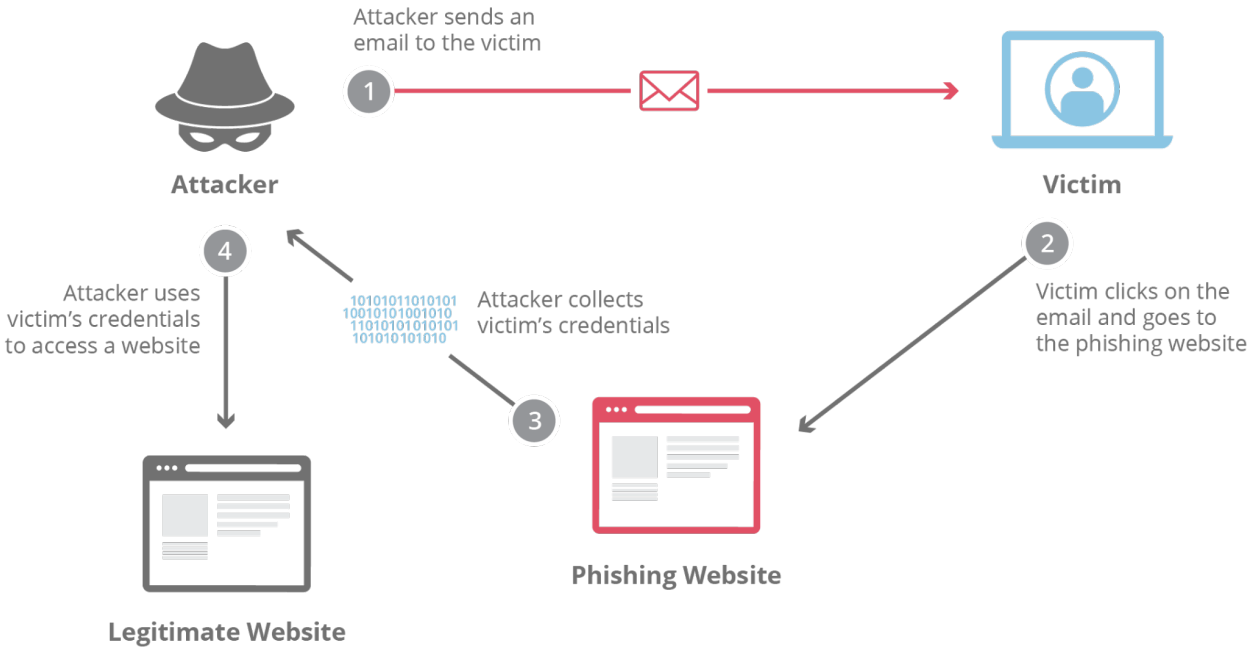
In the current digital era, phishing attacks have become a common and serious cybercrime. Due to the rising importance of online connections and transactions in our daily lives, individuals and organisations are vulnerable to these deceitful strategies. Phishing is an unethical activity in which criminals act as legitimate organisations in an effort to deceive unwitting victims into disclosing private information or committing harmful acts.

A phishing attack frequently involves the creation of false emails, texts, or websites that closely resemble trustworthy ones. These fake communications frequently use fear, urgency, or attractive incentives to convince recipients to act immediately. Multiple types of sensitive data, such as login credentials, credit card numbers, social security numbers, and others, might be the target of phishing attacks.

Cybercriminals can use victims' unintentional disclosure of their personal information for illegal activities, including identity theft, financial fraud, or illegal access to sensitive systems. Furthermore, phishing attacks can spread malware or ransomware, compromising the security of individuals and organisations.

It takes awareness, knowledge, and technological measures to prevent a phishing attack. People should use caution when opening attachments or clicking links in dubious emails. The risk can be considerably decreased by checking the reliability of websites, using strong passwords, and setting on two-factor authentication. Organisations should hold regular training sessions that educate staff about phishing tactics and create an environment of cybersecurity awareness.

In conclusion, phishing attacks present a serious risk to people and businesses because they focus on human weaknesses to trick users and compromise digital security. We will effectively mitigate the risks these malicious efforts bring by taking proactive actions and promoting a cybersecurity mindset.



Types of Phishing Attacks

Phishing attacks take various forms, using various techniques to trick victims and steal sensitive data. To recognise and protect against these dangers, individuals and organisations must be aware of the various phishing attacks.

1. Email phishing: This kind of phishing attack is the most prevalent, in which criminals send misleading emails that appear to be from legitimate sources like banks, governments, or well-known businesses. The links or attachments in these emails frequently contain malicious links, luring users to click on them and disclose their personal information.
2. Spear phishing: When someone or an organisation is specifically targeted, instead of general email phishing. Attackers use extensive study to customise their communications and make them seem legitimate. Spear phishing attacks increase the likelihood of convincing victims to share sensitive data by tailoring the content and using well-known information.
3. Smishing: This kind of phishing attack uses SMS or text messages to trick victims. Cybercriminals pose as respectable organisations and send communications with harmful links or urgent requests. Smishing attacks try to fool receivers into exposing personal information or downloading malware by taking advantage of the trust and immediate nature of text messaging.

4. Vishing: Voice phishing, also known as "vishing," is the practice of scammers obtaining sensitive data from victims over the phone. Attackers frequently adopt the personas of employees from financial institutions, governmental organisations, or customer service departments to trick victims into exposing personal and financial information over the phone.
5. Pharming: Pharming attacks include secretly redirecting victims from trustworthy websites to malicious ones. This is done either by using malware to change the hosts file on compromised devices or by compromising DNS servers. Cybercriminals can intercept and use any information victims enter into these bogus websites.

People and organisations may implement suitable preventive measures and maintain awareness in the face of increasing cyber risks by being aware of these numerous phishing attack types.

Pishing Simulator

A phishing simulator tool is an application of software built to help organisations determine how vulnerable they are to phishing attacks and inform their staff of the risks involved. The program simulates actual phishing scenarios by delivering simulated emails to employees, monitoring their answers, and giving in-depth analytics and reports.

The phishing simulator program aims to simulate the strategies used by actual cybercriminals in a safe and controlled situation. These technologies allow organisations to evaluate the efficacy of their security measures, identify vulnerabilities, and determine the degree of employee awareness and readiness.

Organisations can create customised phishing campaigns by producing realistic phishing emails that resemble those used by attackers using a phishing simulator tool. Organisations can use the tool to select from various templates, edit the content, and customise the campaigns to meet their unique requirements and goals.

The tool monitors the actions of employees who receive phishing emails and records whether they click on links, open attachments, or expose critical information. When an employee falls for the simulated phishing attack, the simulator immediately informs them and offers advice on how to spot and prevent similar attacks in the future.

Employing a phishing simulator tool allows organisations to reinforce best practices for detecting suspicious emails and educate employees on the strategies criminals use. These tools are essential for improving an organisation's overall security posture and reducing the risks brought on by actual phishing attacks.

7phish

The 7phish tool is an innovative, user-friendly solution for conducting phishing simulations in cybersecurity research and training settings. Organizations can use this tool, developed as a shell script program, to evaluate their vulnerability to phishing attacks and improve their defensive measures against this common cyber threat.

Researchers and security experts may imitate phishing campaigns using 7phish's flexible capabilities, enabling it to generate various phishing URLs. The tool's user-friendly interface makes establishing and customising phishing scenarios easier and more accessible to novice and experienced users.

The main objective of 7phish is to create phishing simulations that accurately replicate the strategies used by real attackers. Using these simulations, organisations can efficiently evaluate how attentive, knowledgeable, and responsive their employees are to phishing attacks. This enables customised training programs to inform employees about the risks of phishing and improve their capacity to identify and avoid such attacks.

Additionally, 7phish's efficiency and ease make it a significant tool for researchers, enabling them to collect information and evaluate the efficacy of various defences against phishing attacks. The tool's adaptability and simplicity make it easy to include in current research frameworks and cybersecurity training programs.

This tool is named 7phish because it is possible to generate phishing links from 7 domains. The 7 main domains are,

1. Facebook
2. Instagram
3. Google
4. Linkedin
5. Twitter
6. Reddit
7. Tiktok

```
[::] Select An Attack For Your Victim [::]

[01] Facebook
[02] Instagram
[03] Google
[04] Linkedin
[05] Twitter
[06] Reddit
[07] Tiktok
```

In addition to these 7 domains, there are several other options related to each other. By then, the user can choose the desired templates. The following picture shows the templates related to Facebook.

```
[01] Traditional Login Page
[02] Advanced Voting Poll Login Page
[03] Fake Security Login Page
[04] Facebook Messenger Login Page
```

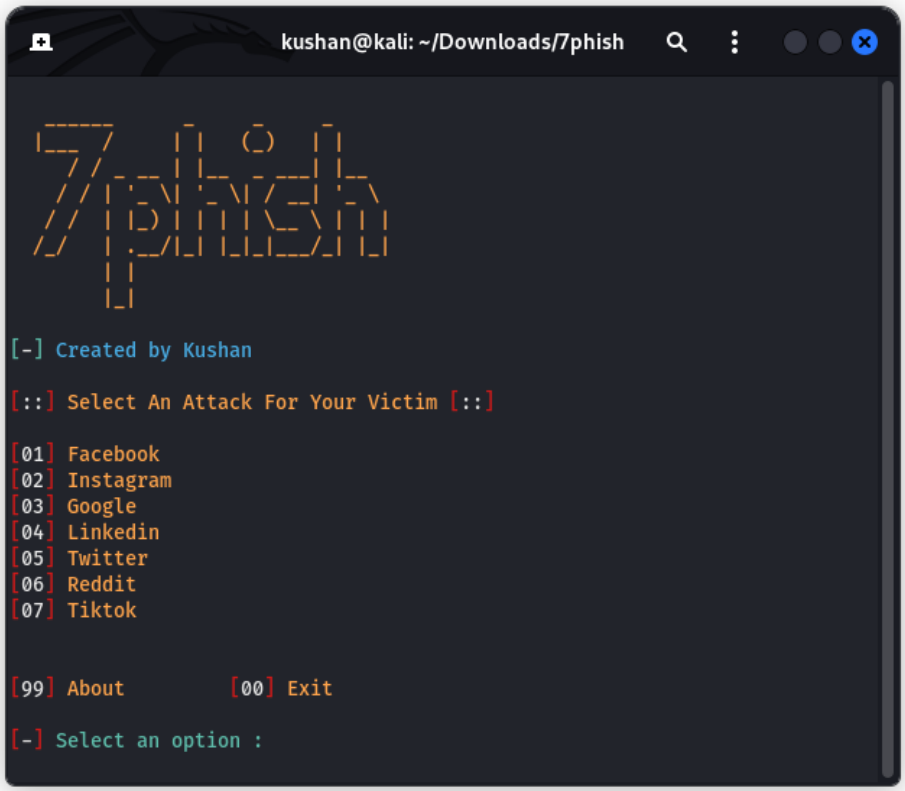
It also includes 3 port forwarding services. The user can also select them as desired. Localhost, Cloudflare, and LocalXpose tunnels have been used for this.

```
[01] Localhost
[02] Cloudflared
[03] LocalXpose
```

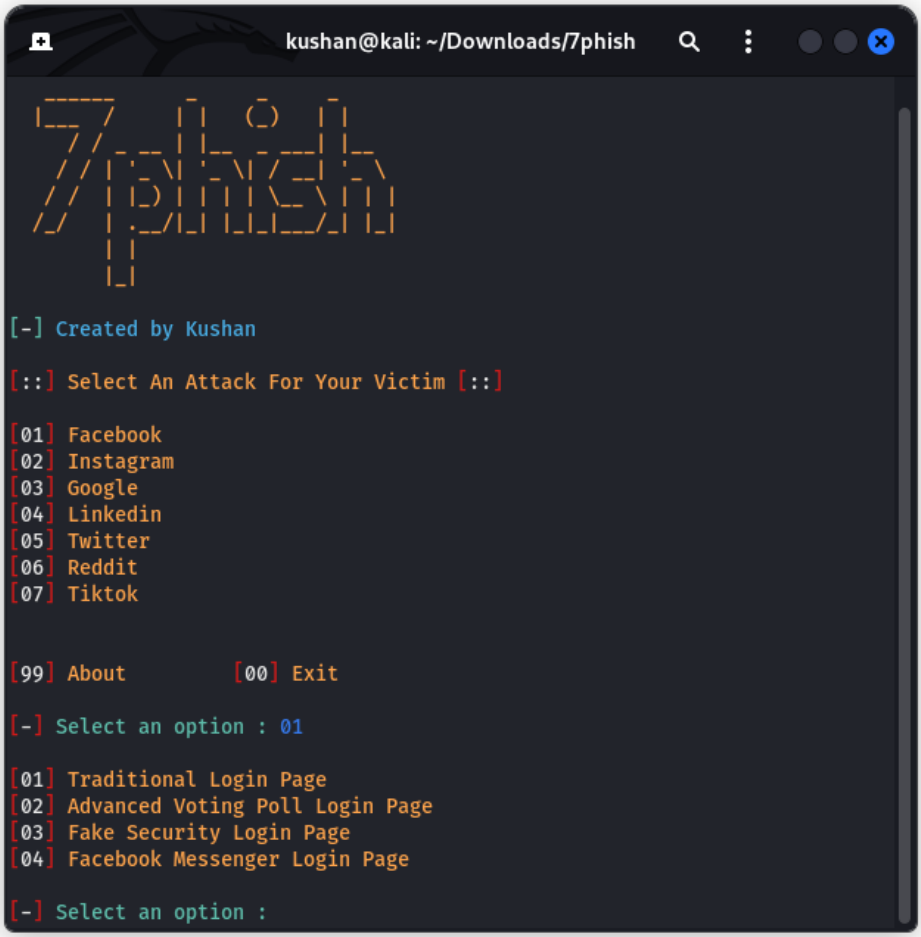
Testing the tool

Here, how to do a phishing simulation using the 7phish tool has been shown step by step.

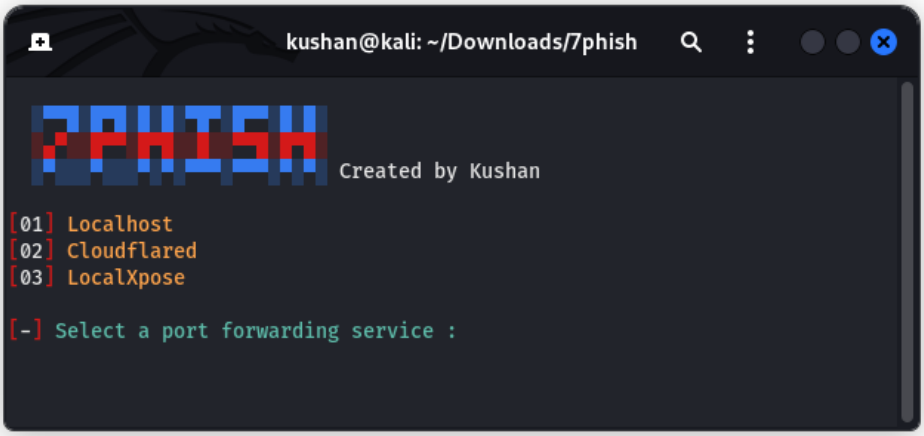
First, go to the directory where the .sh file of this tool is located, open the terminal and enter 'bash 7phish.py' and run the program. Then the 7phish tool will be activated as follows.



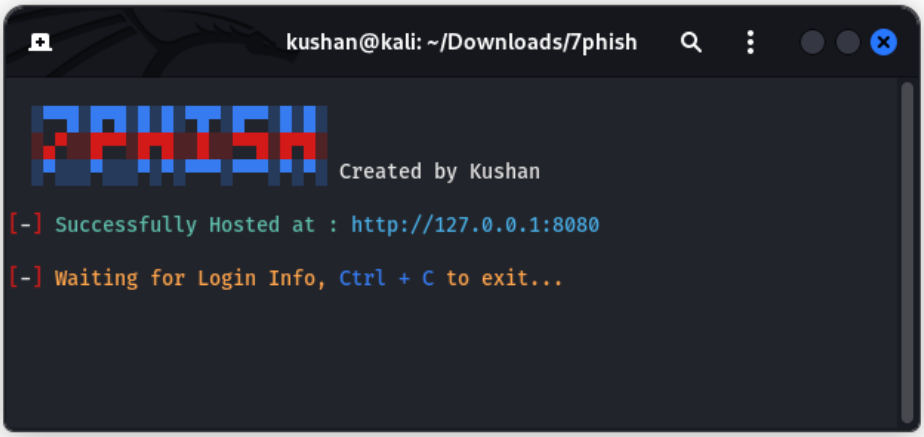
Here, the user can select a desired domain, input the corresponding number and proceed. In this case, Facebook is chosen as an example.



Now select the Traditional Login Page and proceed.

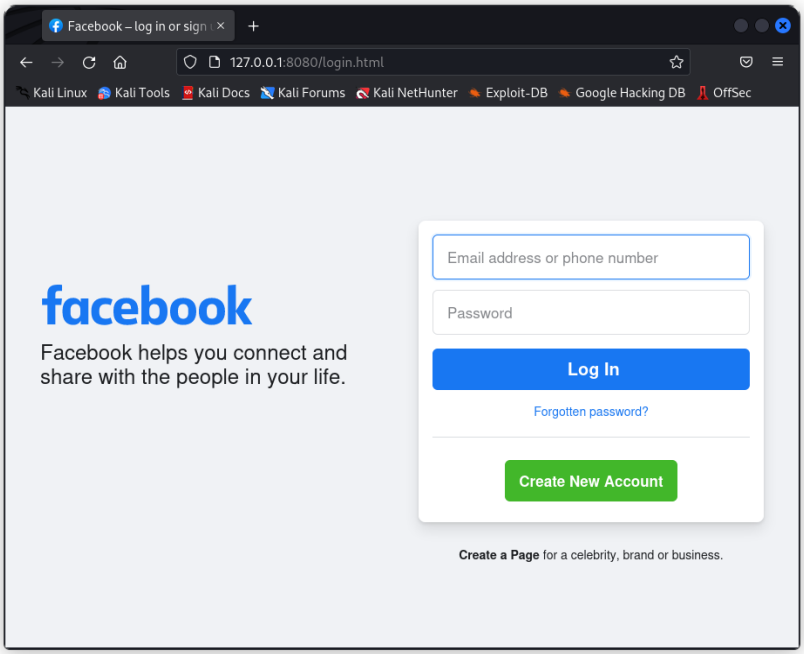


As shown in the picture above, the user can select a Port Forwarding Service as required. Here Localhost is selected as Port Forwarding Service. And here, it asks if you want to customise the port, at this time, answer no and move on.

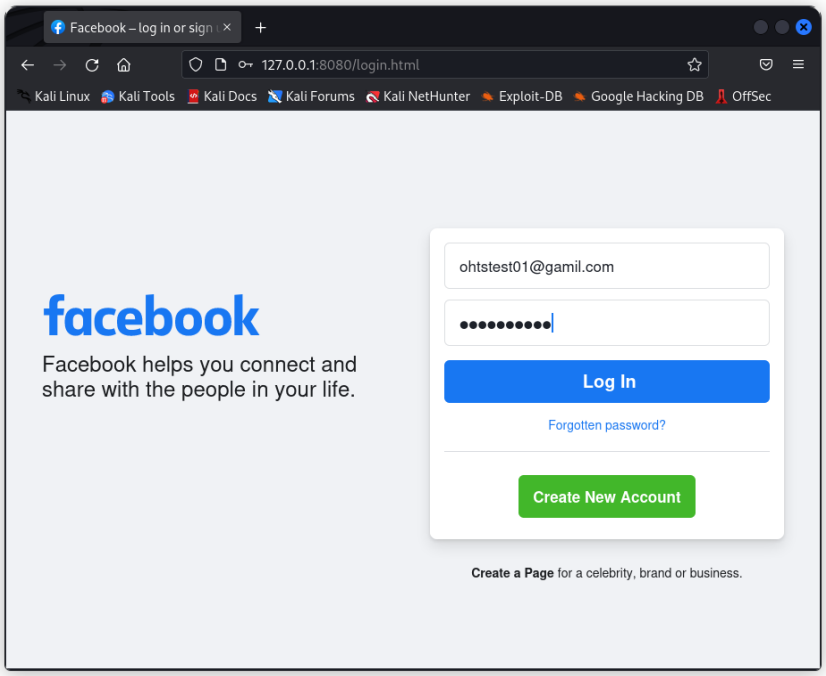


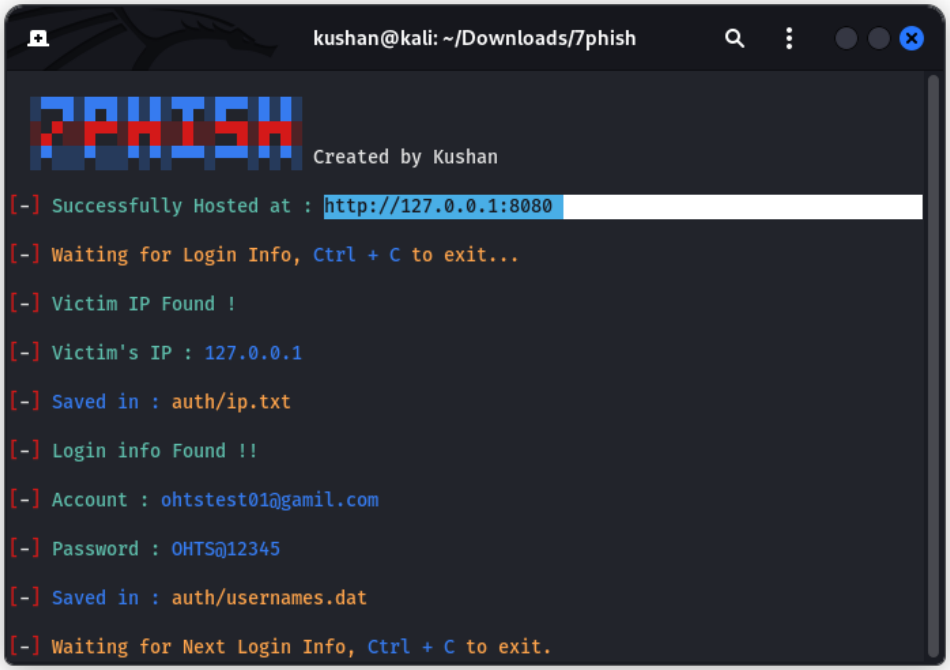
Now the template is prepared as requested by the user and given as the Localhost address.

Since Localhost is taken as the tunnel here, the given Localhost URL should also be tried from a device connected to the Local network. Then the usual Facebook login page will be displayed.



When enter an email and password and click the login button, these entered credentials will be displayed in the 7phish tool.



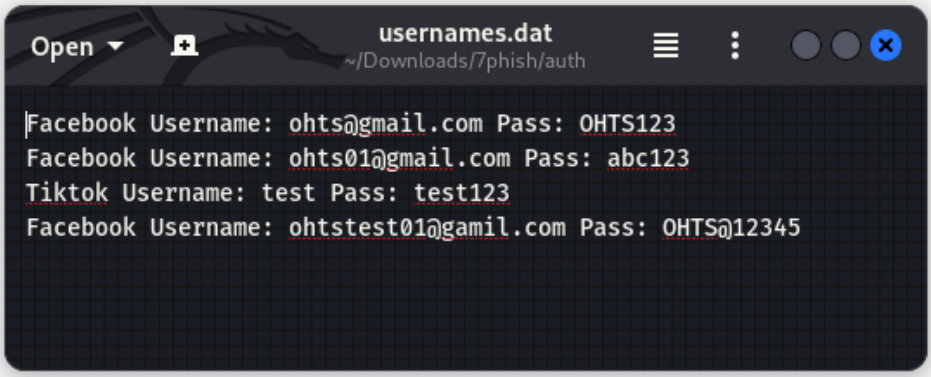


This means that the phishing simulation has been successful.

Results

The information obtained in the abovementioned test is saved in two separate files.

The username/email and password entered by the victim during the phishing simulation are saved in the usernames.dat file.



Details about the IP, OS, Web browser and their versions used by the victim in the phishing simulation are saved in the ip.txt file.

Using this information, employees' knowledge of phishing attacks can be checked, and it also helps to give them good knowledge about these.

Conclusion

The creation of the 7phish tool filled a significant gap in cybersecurity training and research. The tool allows organisations to evaluate their vulnerability to phishing attacks and improve their defence techniques by providing a flexible and user-friendly method for using phishing simulations. Realistic and targeted simulations that properly mimic the strategies employed by real attackers are made possible by the ability to produce different sorts of phishing links and personalised campaigns. Organisations can evaluate employee awareness through these simulations, educate employees on phishing risks and implement preventative measures to prevent attacks in the future. Researchers can also use 7phish to gather important information and evaluate the efficacy of defences, furthering their understanding of cybersecurity. Thanks to its adaptability and simplicity, the 7phish tool is an invaluable resource for organisations and researchers, allowing them to improve their cybersecurity posture and keep up with new phishing threats.

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